Marwadi University, Marwadi Chandarana Group, NAAC A+	**Marwadi University** **Faculty of Engineering & Technology** **Department of Information and Communication Technology**	
Subject: Digital Signal and Image Processing (01CT1513)	**Aim:** Simulate linear convolution and circular convolution on discrete time signals.	
Experiment No: 2	**Date:**	**Enrollment No:92510133011**

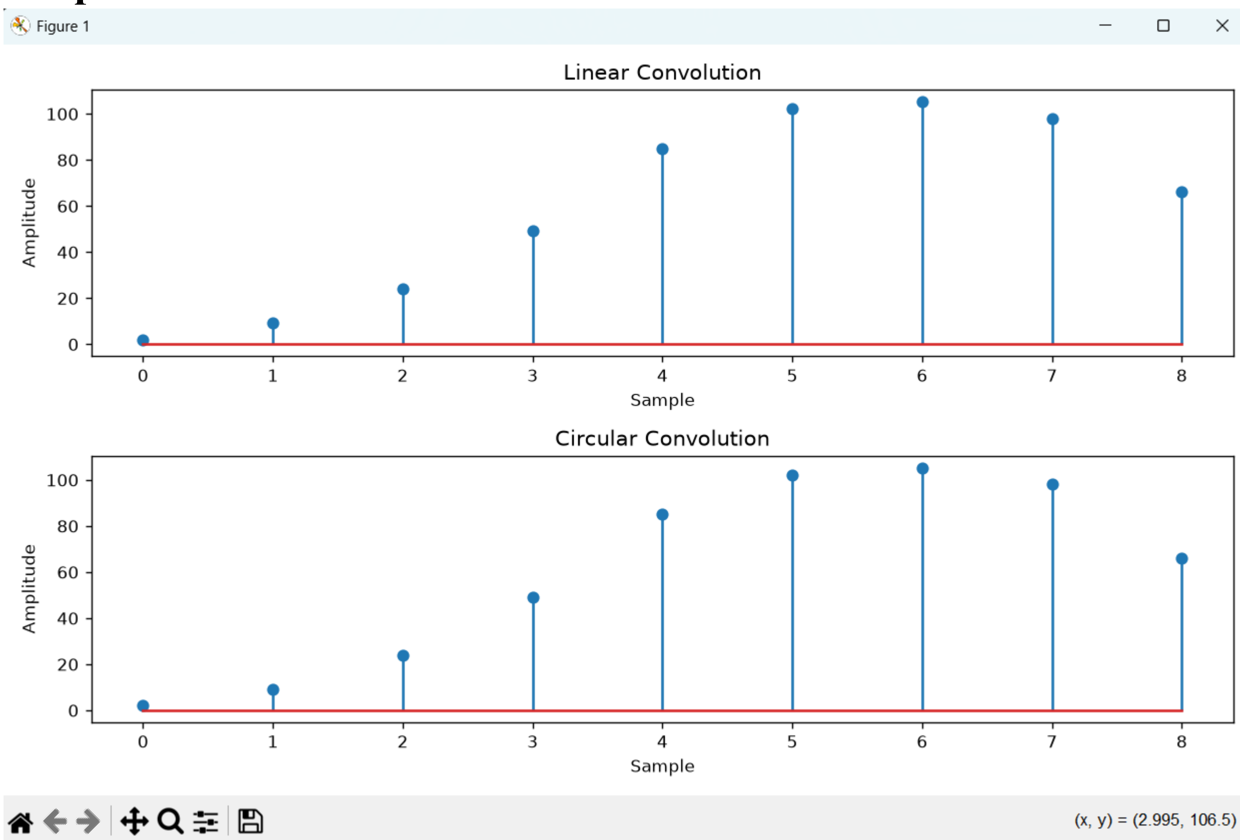
Github:-

https://github.com/trupalijasani05/trupali-jasani

Code:

```
import numpy as np
import matplotlib.pyplot as plt
def linear_convolution(signal1, signal2):
linear_conv = np.convolve(signal1, signal2, mode='full') return
linear_conv
def circular_convolution(signal1, signal2):
    # Compute the circular convolution
    fft_length = len(signal1) + len(signal2) - 1
    fft_signal1 = np.fft.fft(signal1, fft_length)
    fft_signal2 = np.fft.fft(signal2, fft_length)
    circular_conv = np.fft.ifft(fft_signal1 * fft_signal2)
    return circular_conv
# Define the discrete-time signals
signal1 = np.array([2, 3, 3, 4, 6])
signal2 = np.array([1, 3, 6, 9, 11])
# Compute the linear convolution
linear_conv = linear_convolution(signal1, signal2)
# Compute the circular convolution
circular_conv = circular_convolution(signal1, signal2)
# Plot the linear and circular convolution results
plt.figure(figsize=(10, 6))
plt.subplot(2, 1, 1)
plt.stem(linear_conv)
plt.title('Linear Convolution')
plt.xlabel('Sample')
plt.ylabel('Amplitude')
plt.subplot(2, 1, 2)
plt.stem(circular_conv)
plt.title('Circular Convolution')
plt.xlabel('Sample')
plt.ylabel('Amplitude')
plt.tight_layout()
plt.show()
```

Output:



Conclusion:

This experiment shows that circular and linear convolution produce the exact same results if you add enough zeros to the signals.
Because we set the FFT length to 9 instead of 5, we gave the signals enough room so they wouldn't wrap around and overlap with themselves. That is why both graphs in the output look completely identical.

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Page No.
Date

experiment - 2
~~Experiment-2.~~

Linear convolution.

signal 1 = [2, 3, 3, 4, 6]
signal 2 = [1, 3, 6, 9, 11]

	2	3	3	4	6
1	2	3	3	4	6
3	6	9	9	12	18
6	12	18	18	24	36
9	18	27	27	36	54
11	22	33	33	44	66

Diagonal addition.

To find value of the o/p sequence $y(n)$, sum the values along the anti-diagonals.

$y(0) = 2$
$y(1) = 6+3 = 9$
$y(2) = 12+9+3 = 24$
$y(3) = 18+18+9+4 = 49$
$y(4) = 22+27+18+12+6 = 85$
$y(5) = 33+27+24+18 = 102$
$y(6) = 33+36+36 = 105$
$y(7) = 44+54 = 98$
$y(8) = 66$

Final result:

The linear convolution of the two signals is:
$y(n) = [2, 9, 24, 49, 85, 102, 105, 98, 66]$

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Page No.
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circular convolution:-

$N = 5 + 5 - 1 = 9$

$x_1 = [2, 3, 3, 4, 6, 0, 0, 0, 0]$

$x_2 = [1, 3, 6, 9, 11, 0, 0, 0, 0]$

$$\begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 3 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 6 & 3 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 9 & 6 & 3 & 1 & 0 & 0 & 0 & 0 & 0 \\ 11 & 9 & 6 & 3 & 1 & 0 & 0 & 0 & 0 \\ 0 & 11 & 9 & 6 & 3 & 1 & 0 & 0 & 0 \\ 0 & 0 & 11 & 9 & 6 & 3 & 1 & 0 & 0 \\ 0 & 0 & 0 & 11 & 9 & 6 & 3 & 1 & 0 \\ 0 & 0 & 0 & 0 & 11 & 9 & 6 & 3 & 1 \end{bmatrix} \begin{bmatrix} 2 \\ 3 \\ 3 \\ 4 \\ 6 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

$y(0) = (2 \times 1) = 2$

$y(1) = (3 \times 2) + (1 \times 3) = 9$

$y(2) = (6 \times 2) + (3 \times 3) + (1 \times 3) = 24$

$y(3) = (9 \times 2) + (6 \times 3) + (3 \times 3) + (1 \times 4) = 49$

$y(4) = (11 \times 2) + (9 \times 3) + (6 \times 3) + (3 \times 4) + (1 \times 6) = 85$

$y(5) = (11 \times 3) + (9 \times 3) + (6 \times 4) + (3 \times 6) = 102$

$y(6) = (11 \times 3) + (9 \times 4) + (6 \times 6) = 105$

$y(7) = (11 \times 4) + (9 \times 6) = 98$

$y(8) = (11 \times 6) = 66$

$y(n) = [2, 9, 24, 49, 85, 102, 105, 98, 66]$